Unit: Unit 3: Functions
Topic: Operations with Functions

Name: ____________
Date: __________

Quiz (Carolina Reaper)

Q1. If $f(x) = x^2$ and $g(x) = 2x + 1$, what is $(f + g)(x)$?
(A) $x^2 + 2x + 1$
(B) $x^2 + 2x - 1$
(C) $x^2 - 2x + 1$
(D) $x^2 - 2x - 1$
(E) $x^2 + 1$

Q2. Given $f(x) = x^3$ and $g(x) = x - 2$, what is $(f - g)(x)$?
(A) $x^3 - x + 2$
(B) $x^3 + x - 2$
(C) $x^3 - x - 2$
(D) $x^3 + x + 2$
(E) $x^3 - 2$

Q3. For $f(x) = 2x$ and $g(x) = 3x$, what is $(fcdotg)(x)$?
(A) $6x$
(B) $5x$
(C) $4x$
(D) x
(E) $8x$

Q4. A delivery truck travels from city A to city B. If the distance traveled is given by $f(x) = x^2 + 2x$ and the time taken is given by $g(x) = x + 1$, what is the domain of $(fcdotg)(x)$ if x represents the speed?
(A) All real numbers
(B) $xgeq0$
(C) $x > 0$
(D) $xgeq - 2$
(E) $x < -2$

Q5. If $f(x) = x^2 - 1$ and $g(x) = x + 2$, what is $(f/g)(x)$?
(A) $\frac{x^2-1}{x+2}$
(B) $\frac{x^2+1}{x-2}$
(C) $\frac{x^2-1}{x-2}$
(D) $\frac{x^2+1}{x+2}$
(E) $\frac{x-1}{x+2}$

Q6. A shipping company has two routes. Route A's distance is given by $f(x) = x^2 + x$ and Route B's distance is given by $g(x) = x - 1$. What is the domain of $(f + g)(x)$ if x represents the number of deliveries?
(A) $xgeq1$
(B) $x > 0$
(C) All real numbers
(D) $x < -1$
(E) $xgeq - 1$

Q7. Simplify $(f - g)(x)$ if $f(x) = 2x^2$ and $g(x) = x^2 - 1$.
(A) $x^2 + 1$
(B) $x^2 - 1$
(C) x^2
(D) x
(E) $x^2 + 2$

Q8. For $f(x) = x - 2$ and $g(x) = x + 1$, what is $(fcdotg)(x)$?
(A) $x^2 - x - 2$
(B) $x^2 + x - 2$
(C) $x^2 - x + 2$
(D) $x^2 + x + 2$
(E) $x^2 - 2$

Open Ended Questions (FRQ)

Q9. A traffic flow model uses the functions $f(x) = x^2 + 2x$ to represent the distance traveled and $g(x) = x + 1$ to represent the time taken. Find the domain of $(f/g)(x)$ and simplify the expression. Show your work.

Q10. A logistics company has two warehouses. The distance from Warehouse A to the delivery point is given by $f(x) = x^2 - 1$ and from Warehouse B by $g(x) = x - 2$. Determine the expression for $(f + g)(x)$ and state its domain if x represents the number of packages. Show your work.

Chef's Tips

- Chef's Tips
- Ensure students simplify their expressions.
- Check for domain restrictions.